

Machine Learning Introduction

1. Preface

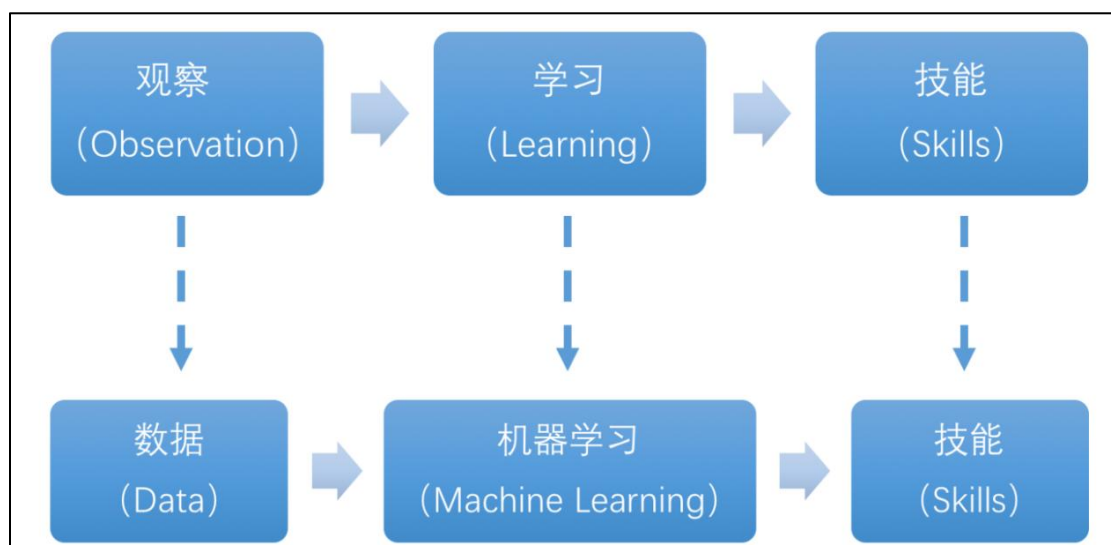
Artificial Intelligence (AI) is a new interdisciplinary field of science and technology that studies and develops theories, methods, techniques, and application systems for simulating, extending, and expanding human intelligence.

AI includes machine learning, computer vision, natural language processing, and deep learning. Machine learning is a subfield of AI. Deep learning is a type of machine learning.

Since its inception, AI has matured in both theory and technology, and its application areas have continued to expand, gradually becoming an independent branch.

2. What “Machine Learning” is

Machine Learning forms the cornerstone of artificial intelligence, serving as the fundamental approach to endow machines with intelligence. It spans multiple interdisciplinary fields such as probability theory, statistics, approximation theory, convex analysis, and algorithm complexity theory.



In essence, machine learning explores how computers can acquire new

knowledge or skills by mimicking human learning behaviors and continuously enhancing their performance by reorganizing existing knowledge structures. Practically, it entails utilizing data to train models and leveraging these models for predictions.

For instance, consider AlphaGo, the pioneering artificial intelligence system that triumphed over human professional Go players and even world champions. AlphaGo operates on the principles of deep learning, wherein it discerns the intrinsic laws and representation layers within sample data to extract meaningful insights.

3. Types of Machine Learning

Machine learning can be broadly categorized into two types: supervised learning and unsupervised learning. The key distinction between these two types lies in whether the machine learning algorithm has prior knowledge of the classification and structure of the dataset.

3.1 Supervised Learning

Supervised learning involves providing a labeled dataset to the algorithm, where the correct answers are known. The machine learning algorithm uses this dataset to learn how to compute the correct answers. It is the most commonly used type of machine learning.

For instance, in image recognition, a large dataset of dog pictures can be provided, with each picture labeled as "dog". This labeled dataset serves as the "correct answer". By learning from this dataset, the machine can develop the ability to recognize dogs in new images.



Model Selection: In supervised learning, selecting the right model to represent the data relationship is crucial. Common supervised learning models encompass linear regression, logistic regression, decision trees, support vector machines (SVM), and deep neural networks. The choice of model hinges on the data's characteristics and the problem's nature.

Feature Engineering: Feature engineering involves preprocessing and transforming raw data to extract valuable features. This encompasses tasks like data cleaning, handling missing values, normalization or standardization, feature selection, and feature transformation. Effective feature engineering can significantly enhance model performance and generalization capabilities.

Training and Optimization: Leveraging labeled training data, we can train the model to capture the data relationship. Training typically involves defining a loss function, selecting an appropriate optimization algorithm, and iteratively adjusting model parameters to minimize the loss function. Common optimization algorithms include gradient descent and stochastic gradient descent.

Model Evaluation: Upon completing training, evaluating the model's performance on new data is essential. Standard evaluation metrics include accuracy, precision, recall, F1 score, and ROC curve. Assessing a model's performance enables us to gauge its suitability for practical applications. In summary, supervised learning entails utilizing labeled training data to train a model for predicting or classifying new unlabeled data. Key steps encompass selecting an appropriate model, conducting feature engineering, training and optimizing the model, and evaluating its performance. Together, these components constitute the foundational elements of supervised learning.

3.2 Unsupervised Learning

Unsupervised learning involves providing an unlabeled dataset to the algorithm, where the correct answers are unknown. In this type of machine learning, the machine must mine potential structural relationships within the dataset.

For instance, in image classification, a large dataset of cat and dog pictures can be provided without any labels. Through unsupervised learning, the machine can learn to divide the pictures into two categories: cat pictures and dog pictures.

